

BOT_ARCHITECTURE.md

Project: Modular AI Trading Engine

Version: 1.0

Philosophy

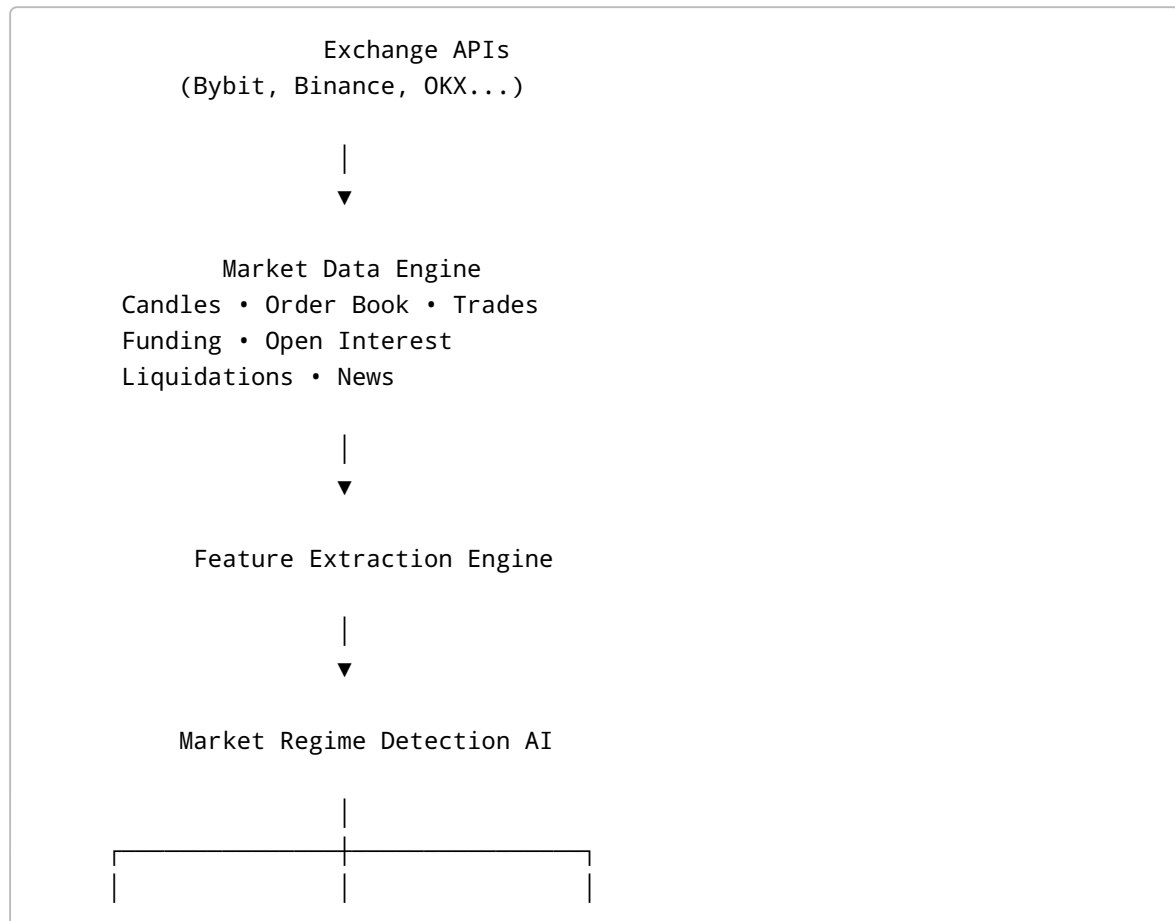
The goal of this project is **not** to create the "best trading strategy."

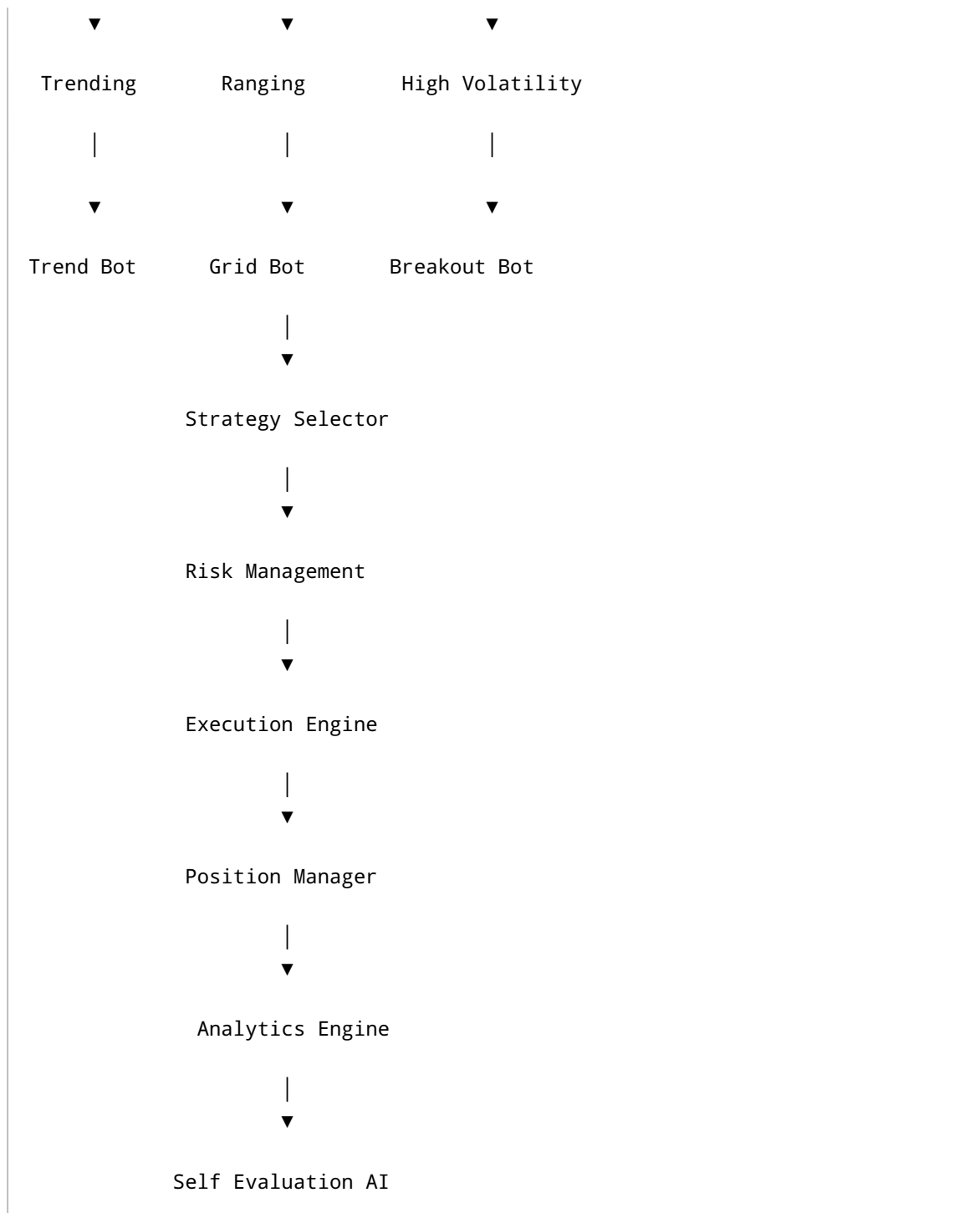
Instead, the objective is to build an intelligent system capable of:

- Understanding the current market
- Selecting the most appropriate strategy
- Managing risk automatically
- Continuously improving through analysis

Think of it as a trading operating system rather than a single bot.

High-Level Architecture





Module 1 — Market Data Engine

Responsible for collecting every piece of information.

Inputs

Market

- OHLCV
- Tick Data
- Trades
- Bid/Ask
- Order Book

Derivatives

- Funding Rate
- Open Interest
- Mark Price
- Liquidation Data

External

- News
 - Fear & Greed Index
 - Economic Calendar
 - Twitter/X
 - Reddit
-

Module 2 — Feature Extraction

Transforms raw data into meaningful signals.

Examples:

Trend

- EMA 20
 - EMA 50
 - EMA 200
 - SMA
 - MACD
 - ADX
-

Volatility

- ATR
 - Bollinger Width
 - Historical Volatility
 - Realized Volatility
-

Momentum

- RSI
 - ROC
 - Williams %R
 - Stochastic
-

Volume

- OBV
 - VWAP
 - Delta
 - Volume Profile
-

Order Flow

- CVD
 - Iceberg Detection
 - Large Trades
 - DOM Imbalance
-

Module 3 — Market Regime Detection

This module determines **what kind of market we're in**.

Possible states:

Trending Up

Characteristics:

- ADX High
 - EMA alignment
 - Higher highs
 - Higher lows
-

Trending Down

Characteristics:

- Lower highs
- Lower lows
- Strong selling pressure

Sideways

Characteristics:

- Low ADX
- Flat EMA
- Tight Bollinger Bands

Volatile

Characteristics:

- ATR Spike
- Wide Bollinger Bands
- Large candles

Compression

Characteristics:

- Very low ATR
- Very low volume

Potential breakout candidate.

News Mode

Detected from:

- CPI
- FOMC
- ETF news
- Major exchange announcements

Module 4 — Strategy Library

Every strategy exists independently.

Examples:

`/strategies`

```
trend/  
grid/  
breakout/  
scalping/  
mean_reversion/  
market_making/  
momentum/  
funding/  
liquidation/  
swing/  
smart_money/
```

Every strategy implements:

```
Analyze()  
  
Entry()  
  
Exit()  
  
StopLoss()  
  
TakeProfit()  
  
Confidence()  
  
Risk()
```

No strategy knows about another strategy.

Module 5 — Strategy Selector

The "brain."

Instead of asking:

Should I buy?

It asks:

Which strategy should control trading right now?

Example:

```
Trend Score = 92
```

Range Score = 18

Breakout Score = 61

Selected Strategy:

Trend Following

Module 6 — Confidence Engine

Every strategy outputs confidence.

Example:

Trend

92%

Grid

34%

Breakout

57%

Momentum

80%

The selector may choose:

- highest confidence
- weighted average
- voting system

Module 7 — Risk Manager

Arguably the most important module.

Controls:

Position Size

Example:

Risk 1% account

Stop = 2%

Position Size = calculated automatically

Dynamic Leverage

Trending:

5x

Volatile:

2x

Range:

3x

Daily Loss

Example

Stop trading after

-3%

Weekly Loss

Stop after

-8%

Max Drawdown

Stop entire system after

15%

Exposure

Never allocate

100%

capital

to one asset.

Module 8 — Execution Engine

Responsible for sending orders.

Supports:

- Limit
- Market
- Post Only
- IOC
- FOK

Handles:

- retries
 - latency
 - slippage
 - partial fills
-

Module 9 — Position Manager

Tracks:

- Active trades
- Unrealized PnL
- Fees
- Funding
- Trailing Stop
- Partial Take Profit

Supports:

- Scaling In
 - Scaling Out
-

Module 10 — Analytics

Logs everything.

Example

Trade ID
BTCUSDT
Trend Strategy
Confidence
89%
Entry
118000
Exit
119800
PnL
+1.8%
Duration
4h
Funding
+0.02%
Slippage
0.01%

Nothing is discarded.

Module 11 — Performance Dashboard

Shows

Daily

Weekly

Monthly

Win Rate

Profit Factor

Sharpe Ratio

Sortino

Drawdown

Average Trade

Largest Win

Largest Loss

Strategy Ranking

Regime Performance

Heatmaps

Module 12 — Self Evaluation

Runs periodically.

Questions:

Which strategy is winning?

Which regime loses money?

Which indicator stopped working?

What hours are most profitable?

What leverage performs best?

Produces optimization reports instead of changing live parameters automatically.

Module 13 — Machine Learning (Optional)

Recommended uses:

- Market regime classification
- Volatility forecasting
- Trade quality scoring
- Position sizing suggestions
- Strategy ranking

Avoid giving the model direct control over order placement without extensive testing and safeguards.

Module 14 — Configuration

Example:

```
config/  
risk.yaml  
strategies.yaml  
exchange.yaml  
telegram.yaml  
discord.yaml  
database.yaml
```

Module 15 — Notifications

Telegram

Discord

Email

Mobile Push

Example:

Opened Long

BTCUSDT

Trend Strategy

Confidence

91%

SL

117200

TP

120500

Suggested Project Structure

src/

core/

market/

strategies/

risk/

execution/

portfolio/

analytics/

database/

exchange/

api/

dashboard/

config/

Development Roadmap

Phase 1

- Exchange connectivity
- Market data ingestion
- Manual order execution

Phase 2

- Basic strategies (Trend, Grid, Mean Reversion)
- Backtesting framework

Phase 3

- Strategy selector
- Risk engine
- Position manager

Phase 4

- Analytics dashboard
- Notifications
- Performance metrics

Phase 5

- Market regime detection
- Confidence engine
- Strategy orchestration

Phase 6

- Machine learning enhancements
- Advanced execution optimization
- Multi-exchange support
- Portfolio optimization

Guiding Principles

- Every module should have a single responsibility.
- Strategies should be interchangeable and independently testable.

- Risk management must be enforced globally, not by individual strategies.
 - Favor explainable decisions over opaque automation.
 - Backtest thoroughly, then validate on paper trading before deploying live.
 - Record everything; today's logs are tomorrow's improvements.
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Long-Term Vision

A modular, exchange-agnostic trading platform that can:

- Trade multiple assets simultaneously.
- Adapt to changing market regimes.
- Compare strategies objectively.
- Scale from a personal trading bot to a multi-strategy portfolio manager.
- Support future additions such as options, AI-assisted analysis, and new exchanges without redesigning the core architecture.